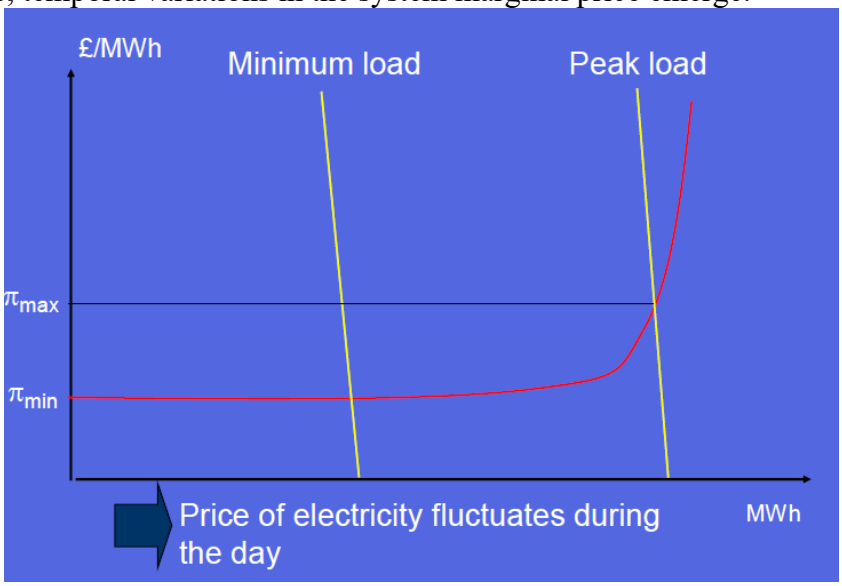
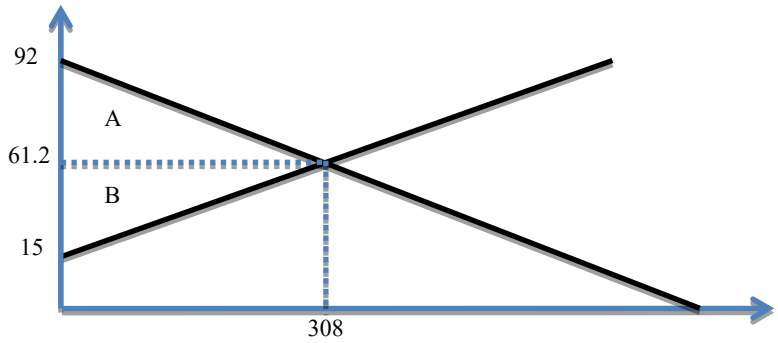
Solutions to Question 1:

(a) The system marginal price signifies the marginal cost of serving an additional unit of demand at the least-cost dispatch of the total load to the available generators. In contrast with other commodities, **electrical energy cannot be economically stored in large quantities** and thus it must be produced exactly when it is consumed by the inflexible demand. Moreover, **the electrical demand is not the same every hour, day, month etc and the generation side is characterized by a production cost which varies with the size of the load.** The combination of these effects explains why the system marginal price varies with time. This is demonstrated in the figure below. The supply curve of the generation side exhibits a "hockey-stick" shape, being rather flat at the area of low loads, where cheap base generators are employed for the satisfaction of the demand, and rather steep at the area of high loads, where expensive peaking generators are also employed to cover the demand. In combination with the fluctuation of the demand from a minimum to a peak value, temporal variations in the system marginal price emerge.



(b)



(b) (i) Inverse demand function:

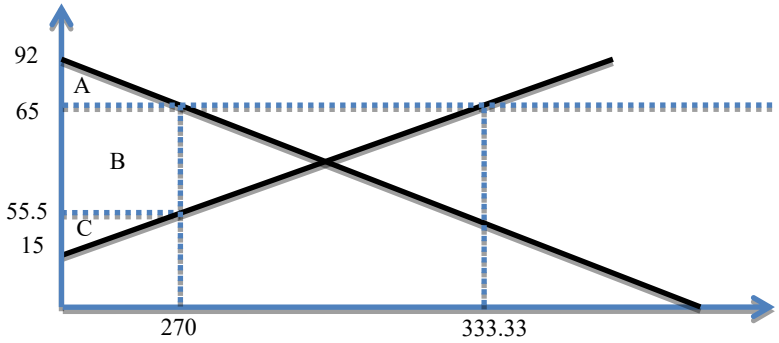
$\pi_D = -0.1Q_D + 92$

(1.1)

Marginal generation cost = inverse supply function:

$\pi_S = 0.15Q_S + 15$
(1.2)
where Q_D, Q_S in [MW] and π_D, π_S in [£/MWh].
Equilibrium is located at the point:
$\pi_s = \pi_D \rightarrow Q_S = Q_D = Q^* = 308\ [MW],\ \ \pi^* = 61.2\ [\frac{£}{MWh}]$
(1.3)
Consumer surplus equal to the area of A = 0.5 × 308 × (92 – 61.2) = 4743.2 [£/h]
(1.4)
Producer surplus equal to the area of B = 0.5 × 308 × (61.2 - 15) = 7114.8 [£/h]
(1.5)
Total social welfare equal to the area of A+B = 4743.2 + 7114.8 = 11,858 [£/h]
(1.6)

(b) (ii)



In this case, the market does not operate at equilibrium because $\pi_D = 65\ [\frac{£}{MWh}]$. For this price, the demand is 270 [MW] and the supply -theoretically- is 333.33 [MW]. However, because electricity supply must equal demand, the supply is 270 [MW] as well. If it was a different commodity, then the extra capacity could be bought by the government and stored for future potential situations of shortage. But this cannot be done with electricity.
Consumer surplus equal to the area of A = 0.5 × 270 × (92 - 65) = 3,645 [£/h]
(1.7)
Producer surplus equal to the area of B+C = 270 × (65 – 55) + 0.5 × 270 × (55.5 – 15) = 8,032.5 [£/h]
(1.8)
Total social welfare equal to the area of A+B+C = 3,645 + 8,032.5 = 11,677.5 [£/h]
(1.9)
Compared to the equilibrium result obtained in (i), it can be observed that the consumption is decreased by $\frac{270-308}{308}$ = 12.34%, the price is increased by $\frac{65-61.2}{61.2}$ = 6.21%, and the producer surplus increased by $\frac{8032.5-7114.8}{7114.8}$ = 12.90% and the consumer surplus reduced by $\frac{3645-4743.2}{4743.2}$ = 23.15%, the total social welfare reduced by $\frac{11677.5-11858}{11858}$ = 1.52%.

This is a case when for example the state intervenes and sets a price floor in order to help the suppliers by allowing them to sell at a higher than the equilibrium price. This policy however results in a reduction of the social welfare.

(c) In the optimal solution of the problem, the marginal cost of each generator is equal to the system marginal cost λ yielding:

$$\frac{\partial C_1(P_1)}{\partial P_1} = 8.05 + 0.4 * P_1 = \lambda \rightarrow P_1 = \frac{\lambda - 8.05}{0.4} \quad (1.10)$$

$$\frac{\partial C_2(P_2)}{\partial P_2} = 8.45 + 0.6 * P_2 = \lambda \rightarrow P_2 = \frac{\lambda - 8.45}{0.6} \quad (1.11)$$

The satisfaction of the system demand implies that:

$$P_1 + P_2 = D = 600\ [MW]$$

(1.12)

Substituting (10)-(11) in (12) gives for the system marginal cost:

$$\frac{\lambda - 8.05}{0.4} + \frac{\lambda - 8.45}{0.6} = 600 \rightarrow \lambda = 152.21\ [\frac{£}{MWh}]$$

(1.13)

Substituting (13) in (10)-(11) yields for the optimal production of each generator:

$$P_1 = 360.40\ [MW]$$

(1.14)

$$P_2 = 239.60\ [MW]$$

(1.15)

Substituting (14)-(15) in the cost functions of the generators yields for the total cost of operating the system:

$$C_{tot} = C_1(P_1) + C_2(P_2) = 48{,}990.92\ [£/h]$$

(1.16)

The system average cost is:

$$\alpha = \frac{C_{tot}}{P_1 + P_2} = 81.65\ [\frac{£}{MWh}]$$

(1.17)

The profit for each generator Ω_i is given by the difference between its respective revenue and cost:

$$\Omega_1 = \lambda * P_1 - C_1(P_1) = 25{,}677.63\ [£/h]$$

(1.18)

$$\Omega_2 = \lambda * P_2 - C_2(P_2) = 16{,}657.45\ [£/h]$$

(1.19)

(d) Given that $P_1 > P_1^{max}$ the solution calculated in **(c)** is not feasible when taking into account given maximum output limits of the generators. The optimal solution can now be calculated by fixing the output of generator 1 to:

$$P_1' = P_1^{max} = 350\ [MW]$$

(1.20)

and generator 2 satisfying the remaining demand and also is feasible by its maximum output limit 300MW:

$$P_2' = D - P_1' = 250\ [MW]$$

(1.21)

Thus, the generators 2 is the marginal unit. Following the equation as in (c), we get:

$\frac{\lambda'-8.45}{0.6} = 250 \rightarrow \lambda' = 158.45\ [\frac{£}{MWh}]$
(1.22)
Substituting (20) and (21) in the cost functions of the generators yields for the total cost of operating the system:
$C'_{tot} = C_1(P'_1) + C_2(P'_2) = 49{,}045.00\ [£/h]$
(1.23)
The system average cost is:
$a' = \frac{C'_{tot}}{P'_1+P'_2} = 81.47\ [\frac{£}{MWh}]$
(1.24)
The profit for each generator Ω_i is given by the difference between its respective revenue and cost:
$\Omega'_1 = \lambda * P'_1 - C_1(P'_1) = 27{,}840.00\ [£/h]$ (1.25)
$\Omega'_2 = \lambda * P'_2 - C_2(P'_2) = 18{,}185.00\ [£/h]$ (1.26)
When the maximum output limits are considered, the output of generator 1 needs to be restricted to a lower value and therefore the output of the generator 2 is increased. Given that generator 2 is characterised by higher variable cost, this new dispatch is more costly (i.e., it increases the total cost of operating the system, the system marginal cost and the average system cost). On the other hand, this new dispatch increases the profit of both the generators. For generators 2 this is driven by the increase of both its outputs and the system marginal price, while for generator 1 this is driven by the increase of the system marginal price despite the reduction of its output.

Solutions to Question 2:

(a) A brief discussion of the following points is expected:
1) There tend to be fairly large participants able to influence the energy price by their actions.
2) The low elasticity of demand facilitates the abuse of market power.
3) Locational effects could prevent the free flow of the commodity.
4) Electricity networks are natural monopolies.

(b) (i) In the Bertrand competition model, Company *B* will set its price marginally below the generation cost of Company *A* in order to squeeze it out of the market. The market price therefore be:
$\pi = 75\left[\frac{£}{MWh}\right]$
(2.1)
The demand level for this price will be:
$D = \frac{250-\pi}{2} = 87.5\left[\frac{£}{MWh}\right]$
(2.2)
Then, all the demand will be supplied by Company B.
$P_A = 0[MW]$
(2.3)

$P_B = 87.5[MW]$
(2.4)
Finally, the profit made by companies A and B will be:
$\Omega_A = (\pi - 75) * P_A = 0[\frac{£}{h}]$
(2.5)
$\Omega_B = (\pi - 62) * P_B = 1137.5[\frac{£}{h}]$
(2.6)

(b) (ii) In the Cournot model of competition, the state of the market is determined by the production decisions made by each firm. We summarize the possible outcomes using a table where all the cells in a column corresponds to a given production by A and the cells in a row corresponds to a given production by company B.
Each cell contains four pieces of information arranged in the following format:

D	Ω_A
Ω_B	π

where:
π: price $\left[\frac{£}{MWh}\right]$
D: demand $[MW]$
Ω_A: profit made by company A $[\frac{£}{h}]$
Ω_B: profit made by company B $[\frac{£}{h}]$
Given the productions P_A and P_B of the two companies, the other quantities are calculated as follows:
$D = P_A + P_B$
(2.7)
$\pi = 250 - 2 * D$
(2.8)
$\Omega_A = P_A * (\pi - 75)$
(2.9)
$\Omega_B = P_B * (\pi - 62)$
(2.10)
Based on the above, the table expressing the Cournot model of competition for the conditions of the problem is shown below. A Nash equilibrium implies that given the production level of either of the two companies in this state, the other company cannot achieve a higher profit by allowing a production level different than the one at the equilibrium. It can thus be observed form the table below that the cell corresponding to $P_A = 30MW$ and $P_B = 30MW$; $P_A = 25MW$ and $P_B = 35MW$ are two Nash equilibria.

P_B/P_A	20		25		30		35	
25	45	1700	50	1875	55	1950	60	1925
	2450	160	2200	150	1950	140	1700	130
30	50	1500	55	1625	60	1650	65	1575
	2640	150	2340	140	2040	130	1740	120
35	55	1300	60	1375	65	1350	70	1225

	2730	140	2380	130	2030	120	1680	110
40	60	1100	65	1125	70	1050	75	875
	2720	130	2320	120	1920	110	1520	100

(c) The cell corresponding to $P_A = 20MW$ and $P_B = 25MW$ involves higher profits than the two cells corresponding to the Nash Equilibria: $P_A = 30MW$ and $P_B = 30MW$; $P_A = 25MW$ and $P_B = 35MW$ (identified as equilibria in **(b)**). However, this cell does not satisfy the properties of NE since: a) given that $P_A = 20MW$, company B can achieve higher profits by producing $P_B = 35MW$ ($\Omega_B = 2{,}730\,£/h$) than $P_B = 25MW$ ($\Omega_B = 2{,}450\,£/h$) and also b) given that $P_B = 25MW$, company A can achieve higher profits by producing $P_A = 30MW$ ($\Omega_A = 1{,}950\,£/h$) than $P_A = 20MW$ ($\Omega_A = 1{,}700\,£/h$).

(d) Each generating company is trying to maximize its profit which is equal to the difference between its revenue and its cost.

$$\Omega_A = P_A * \pi(D) - C_A(P_A)$$
(2.11)

$$\Omega_B = P_B * \pi(D) - C_B(P_B)$$
(2.12)

However, we cannot treat these two maximizations as separate optimization problems since they are links through the demand:

$$D = P_A + P_B$$
(2.13)

$$\pi = 250 - 2 * D$$
(2.14)

We can around this difficulty by expressing the price as a function of the production of the two generators:

$$\pi = 250 - 2 * (P_A + P_B)$$
(2.15)

The profit of the two companies than be expressed as:

$$\Omega_A = P_A * (250 - 2 * P_A - 2 * P_B) - 75 * P_A = -2 * P_A^2 + (175 - 2 * P_B) * P_A$$
(2.16)

$$\Omega_B = P_B * (250 - 2 * P_A - 2 * P_B) - 62 * P_B = -2 * P_B^2 + (188 - 2 * P_A) * P_B$$
(2.17)

The optimality condition for the maximization problem of company A is:

$$\frac{\partial \Omega_A}{\partial P_A} = -4 * P_A - 2 * P_B + 175 = 0$$
(2.18)

The optimality condition for the maximization problem of company B is:

$$\frac{\partial \Omega_B}{\partial P_B} = -4 * P_B - 2 * P_A + 188 = 0$$
(2.19)

(We only consider the partial derivative of the profit of a generator with respect to its own output because that is the only variable over which it has control)

Putting these optimality conditions in matrix form we get:

$$\begin{bmatrix} -4 & -2 \\ -2 & -4 \end{bmatrix} \begin{bmatrix} P_A \\ P_B \end{bmatrix} = \begin{bmatrix} -175 \\ -188 \end{bmatrix}$$
(2.20)

Solving this linear system we get the following productions for the NE:

$P_A = 27[MW]$

(2.21)

$P_B = 33.5[MW]$

(2.22)

Form which we can easily find:

$\pi = 129\left[\frac{£}{MWh}\right]$, $D = 60.5\ [MW]$, $\Omega_A = 1{,}485[\frac{£}{h}]$, $\Omega_B = 2{,}244.5[\frac{£}{h}]$

(2.23)

Solutions to Question 3

(a) In the unconstrained case marginal costs of all active generators should be the same. Unit C has a constant marginal cost which does not depend on its output level. If units A and B reached the output corresponding to 45 £/MWh, we would have:

$$12 + 0.1PA = 45 \rightarrow PA* = 330\,MW$$

(3.1)

$$18 + 0.2PB = 45 \rightarrow PB* = 135\,MW$$

(3.2)

This suggests that the total output would need to be at least 465 MW in order for the marginal cost to reach 45 £/MWh, and the total load in our three-bus system is only 300 MW. Hence, the marginal cost will be lower than 45 £/MWh and unit C will not be used. Dispatch of units A and B is found from the following system of equations:

$$MCA = MCB \rightarrow 12 + 0.1PA = 18 + 0.2PB$$

(3.3)

$$PA + PB = 240 + 60 = 300$$

(3.4)

This yield: PA = 220 MW, PB = 80 MW, PC = 0 MW

All nodal prices are equal to:

$$\pi 1 = \pi 2 = \pi 3 = 12 + 0.1 \cdot 220 = 34\,£/MWh$$

(3.5)

The cost functions of the two units are found by integrating the marginal cost functions (assuming the constant to be zero):

$$CA(PA) = 12PA + 0.05PA^2$$

(3.6)

$$CB(PB) = 18PB + 0.1PB^2$$

(3.7)

The hourly costs are therefore: CA = 5060 £/h, CB = 2080 £/h. And the total hourly cost is 7140 £/h.

(b) The line flows are obtained from the following system of linear equations:

$$F12 + F13 = PA = 220$$

(3.8)

$$F23 - F12 = PB - DB = 20$$

(3.9)

$$F12 \cdot X12 + F23 \cdot X23 - F13 \cdot X13 = 0$$

(3.10)

The solution to this system is the following: F_{12} = 50 MW, F_{23} = 70 MW, F_{13} = 170 MW.

(c) With the 160 MW limit on line 1-3, we have a 10 MW overload in the unconstrained case. In order to relieve the overloading, it is necessary to decrease output from unit A and increase from unit B by the same amount. The change in outputs needs to result in a decrease of flow in line 1-3 by 10 MW. It will depend on the network topology, as specified in the following system of equations:

$$\Delta P_A + \Delta P_B = 0$$
(3.11)
$$\Delta P_A \cdot \frac{X_{12}+X_{23}}{X_{12}+X_{23}+X_{13}} + \Delta P_B \cdot \frac{X_{23}}{X_{12}+X_{23}+X_{13}} = 0.75\Delta P_A + 0.25\Delta P_B = -10$$
(3.12)
The solution to this system is: $\Delta P_A = -20$ MW, $\Delta P_B = 20$ MW.
This means that the new dispatch will be: $P_A = 200$ MW, $P_B = 100$ MW, $P_C = 0$ MW.
Line loadings for this case will be: $F_{12} = 40$ MW, $F_{23} = 80$ MW, $F_{13} = 160$ MW.

(d) The hourly cost of this dispatch, using the same approach as in a), is: $C_A = 4400$ £/h, $C_B = 2800$ £/h, and the total hourly cost is 7200 £/h.
The hourly cost of security is the difference between generation costs in constrained and unconstrained case: $C_{sec} = 7200 - 7140 = 60$ £/h.

(e) Marginal cost of electricity at every node is the cost of supplying an additional MWh at that node. Marginal generator at bus 1 is unit A, and its marginal cost is:
$$\pi 1 = 12 + 0.1 \cdot 200 = 32\,£/MWh$$
(3.13)
Marginal generator at bus 2 is unit B, as unit A cannot increase its output due to the flow limit on line 1-3. Therefore, nodal price at bus 2 is:
$$\pi 2 = 18 + 0.2 \cdot 100 = 38\,£/MWh$$
(3.14)
For the bus 3, we cannot simply increase outputs of units A or B by 1 MW as this would overload the line 1-3. Due to network topology, an increase of output of unit A by 1 MW causes an overloading of line 1-3 by 0.75 MW, while the same increase by unit B causes an overload of 0.25 MW. Therefore, in order to supply 1 MW of electricity to bus 3 without increasing the loading of line 1-3, we need to solve the following equations:
$$\delta PA + \delta PB = 1$$
(3.15)
$$0.75 \cdot \delta PA + 0.25 \cdot \delta PB = 0$$
(3.16)
Solving this yield: $\delta P_A = -0.5$ MW, $\delta P_B = 1.5$ MW, which means that unit A needs to reduce its output by 0.5 MW, and unit B needs to increase output by 1.5 MW in order to supply additional 1 MW to bus 3.
The nodal price for bus 3 is therefore:
$$\pi 3 = 1.5MCB - 0.5MCA = 41\,£/MWh$$
(3.17)
An alternative approach would be to supply the additional 1 MW by unit C, but its marginal cost is higher than the above price, so it will not be used.
The sum of congestion surpluses across all three lines is obtained as follows:
$$Surplus = (\pi 2 - \pi 1)F12 + (\pi 3 - \pi 2)F23 + (\pi 3 - \pi 1)F13 = 1920\,£/h \qquad (3.18)$$

Solution to Question 4

(a) A brief discussion of the following points is expected:
Transmission is a natural monopoly Because of their visual impact on the environment, construction of competing transmission corridors along similar routes is not feasible. Like all monopolies that provide an essential service, electricity transmission must be regulated to ensure that it delivers an economically optimal combination of quality of service and price.
Transmission is a capital-intensive business The cost of investment in transmission is significantly higher that cost of operating the system. Making efficient investment decisions is thus critical aspect of running a transmission company.
Transmission assets have a long life Most transmission equipment is designed for an expected life of more than 40 years. Generation and demand conditions can change over such a long period. A transmission line that is built on the basis of erroneous forecasts may therefore be used at only a fraction of its rating.
Transmission investments are irreversible Once a transmission line has been built, it cannot be redeployed cost effectively to another location - resale value of installed assets is very low. Owners of transmission networks therefore need to live with the consequences of their investment decisions for a very long time. A large investment that is not used as much as was initially expected is called a *stranded investment*.
Transmission investments are lumpy Manufacturers sell transmission equipment in only a small number of standardized voltage and MVA ratings. It is therefore often not possible to build a transmission facility with rating that exactly matches the need.
Economies of scale Because of high fixed cost component of building transmission infrastructure, the average cost of transmitting electricity generally decreases with the amount transported.

(b) (i) When the capacity of the transmission link is not binding, the optimal generation dispatch is calculated as if both generators and both loads are connected to the same bus and is given by the solution of the system of equations:

$$\frac{\partial C_A(P_A^{unc})}{\partial P_A} = \frac{\partial C_B(P_B^{unc})}{\partial P_B} \rightarrow 10 + 0.02 * P_A^{unc} = 5 + 0.01 * P_B^{unc}$$

(4.1)

$$P_A^{unc} + P_B^{unc} = D_A + D_B \rightarrow P_A^{unc} + P_B^{unc} = 1900$$

(4.2)

which yields:

$$P_A^{unc} = 466.667\ [MW]$$

(4.3)

$$P_B^{unc} = 1433.333\ [MW]$$

(4.4)

Since $P_A^{unc} < D_A$ and $P_B^{unc} > D_B$, the flow on the transmission link is from area B to area A and is equal to:

$$F^{unc} = D_A - P_A^{unc} = P_B^{unc} - D_B = 1033.333\ [MW]$$

(4.5)

The locational marginal price is the same for the two areas and is given by:

$$\pi = 10 + 0.02 * P_A^{unc} = 5 + 0.01 * P_B^{unc} = 19.333\ [\frac{£}{MWh}]$$

(4.6)

The total cost of operating the system is calculated by setting the values determined above in the cost functions of the two generators:

$$C_{tot}^{unc} = C_A(P_A^{unc}) + C_B(P_B^{unc}) = 25783.333\ [£]$$

(4.7)

Since the capacity of the transmission link is not binding, the cost of constraints is zero. The congestion surplus is also zero, since the locational marginal price is the same for the two areas.

(ii) When no transmission link exists between the two areas, generator A will satisfy demand in area A and generator B will satisfy demand in area B, meaning:

$$P_A^{no} = D_A = 1500\ [MW]$$

(4.8)

$$P_B^{no} = D_B = 400\ [MW]$$

(4.9)

The locational marginal prices in the two areas are different. An additional unit of demand in Area A will be satisfied by Generator A and thus the locational marginal price at area A is:

$$\pi_A = \frac{\partial C_A(P_A^{no})}{\partial P_A} = 40\ [\frac{£}{MWh}]$$

(4.10)

An additional unit of demand in Area B will be satisfied by Generator B and thus the locational marginal price at area B is:

$$\pi_B = \frac{\partial C_B(P_B^{no})}{\partial P_B} = 9\ [\frac{£}{MWh}]$$

(4.11)

The total cost of operating the system is calculated by setting the values determined by above in the cost functions of the two generators:

$C_{tot}^{no} = C_A(P_A^{no}) + C_B(P_B^{no}) = 41800$ [£]
(4.12)
The cost of constraints is equal to the difference between the total cost of operating the system in this case and the respective cost in the case where the capacity of the transmission link is not binding:
$CC^{no} = C_{tot}^{no} - C_{tot}^{unc} = 16016.67$ [£]
(4.13)
The congestion surplus is zero since the capacity of the transmission link connecting the two areas is also zero.

(c) (i) The demand function for transmission is calculated as the price differential between the two areas as a function of the line capacity F:
$\Pi_D(F) = \pi_A(F) - \pi_B(F)$
(4.14)
The price at each area depends on the output of the respective generator:
$\pi_A(F) = (10 + 0.02 * P_A^F)$
(4.15)
$\pi_B(F) = (5 + 0.01 * P_B^F)$
(4.16)
The generators' outputs depend on F:
$P_A^F = D_A - F$
(4.17)
$P_B^F = D_B + F$
(4.18)
By substituting (4.15)-(4.18) in (4.14) we get:
$\Pi_D(F) = 31 - 0.03 * F$
(4.19)

(ii) The supply function for transmission is given by the marginal cost of building the transmission line, expressed in an hourly basis:
$\Pi_S(F) = \frac{k*L}{8760h} = 10.274 \left[\frac{£}{MWh}\right]$
(4.20)

(iii) The congestion surplus is given by:
$CS(F) = \left(\pi_A(F) - \pi_B(F)\right) * F$
(4.21)
By substituting (4.19) into (4.21) we get:
$CS(F) = 31 * F - 0.03 * F^2$
(4.22)

(d) (i) A regulated transmission company seeks to maximise the social welfare, by building a line that will exactly finance itself, meaning that the annual revenues from the congestion of the line are equal to the annual investment required for building the line. Therefore, the optimal capacity that it would build satisfies the equality between the demand and the supply for transmission:

$$\Pi_D(F) = \Pi_S(F) \rightarrow 31 - 0.03 * F = 10.274 \rightarrow F = 690.868\ [MW]$$
(4.23)

(ii) A merchant transmission company seeks to maximize its profits. Since its revenues will be given by the congestion surplus -which is equal to the product of the volume of energy transported and the price differential between the two areas- it has an incentive to under-invest in order to keep the price differential in a high level. Therefore, the optimal capacity that it would build maximizes its profits:

$$\max_F(REVENUE(F) - COST(F))$$

Its revenue is equal to the congestion surplus which is given by:

$$REVENUE(F) = CS(F) = 31 * F - 0.03 * F^2$$

Its cost is given by the cost of reinforcing the transmission line, expressed in an hourly basis, which is:

$$COST(F) = 10.274 * F$$

The above yield:

$$\max_F(31 * F - 0.03 * F^2 - 10.274 * F) \rightarrow F = 345.434\ [MW]$$
(4.24)